

Analysis and Design of LDO and OTA on Cadence Virtuoso

EECG221

Team 1

Supervised by: **Eng. Ahmed Amr**

Name	Section No.	ID
Ahmed Wael Mohammed	1	91240930
Adham Mohamed elSadiq	1	91240142
Mohamed Sameer AbdelHamid	3	91240662

1 OTA Design

1.1 Required Specs

Technology	65nm CMOS
Supply Voltage	1.2 V
Open loop DC voltage gain	$\geq 34\text{dB}$
CM input range – low	$\leq 0.65\text{ V}$
CM input range – high	$\geq 1\text{ V}$
GBW	$\geq 5\text{MHz}$
Phase margin	$\geq 60^\circ$
CMRR @ DC	$\geq 45\text{ dB}$

1.2 Design

1.2.1 Design of Input Pair

From the design specs, We notice that the open loop DC gain is 34 dB or roughly 50 V/V , so we used only one stage.

$$GBW = \frac{gm}{2 \cdot \pi \cdot C_L}$$

Let's assume that C_L is 2 pF .

$$gm_{0,1} = 5 \cdot 10^6 \cdot 2 \cdot \pi \cdot C_L = 62.8\mu\text{S}$$

Let us assume that gm is $75\mu\text{S}$ to have some margin for parasitic capacitances.

As shown on the schematic in Figure 1, We need large gm/I_d for $M0$ and $M1$

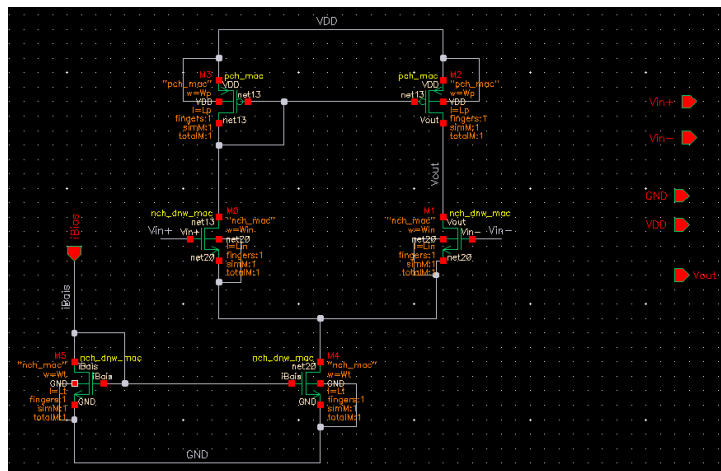


Figure 1: Schematic of Single-Stage OTA

Assume gm/I_d of 15 S/A so,

$$i_d = \frac{gm_{0,1}}{i_d} = 5\mu A$$

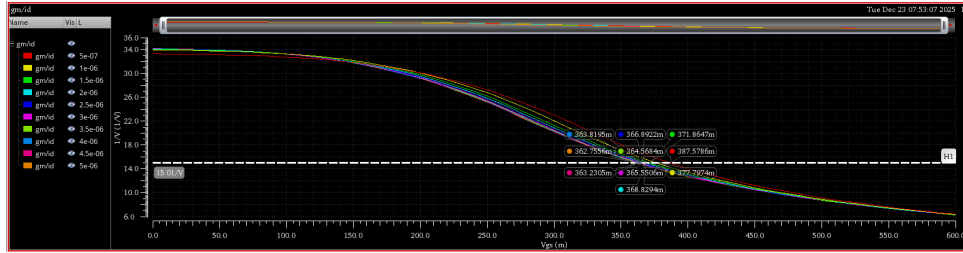


Figure 2: $\frac{gm}{I_d}$ vs V_{GS} with sweeping channel length

Taking V_{GS} of 0.37 V As shown in Figure 2, We want

$$\frac{gm}{2 \cdot g_{ds}} > 50$$

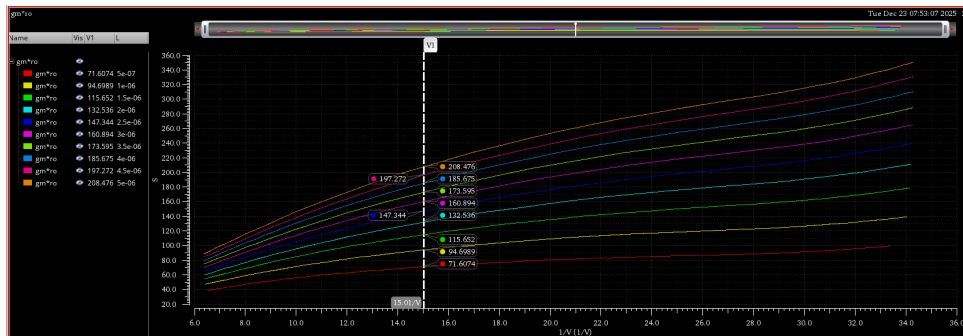


Figure 3: $\frac{gm}{g_{ds}}$ vs $\frac{gm}{I_d}$

so $L > 1.5\mu m$

As shown in Figure 4, for $L = 2\mu m$ and i_d of $5\mu A$ so, at i_d/W of 1.02

$$W = 5\mu m$$

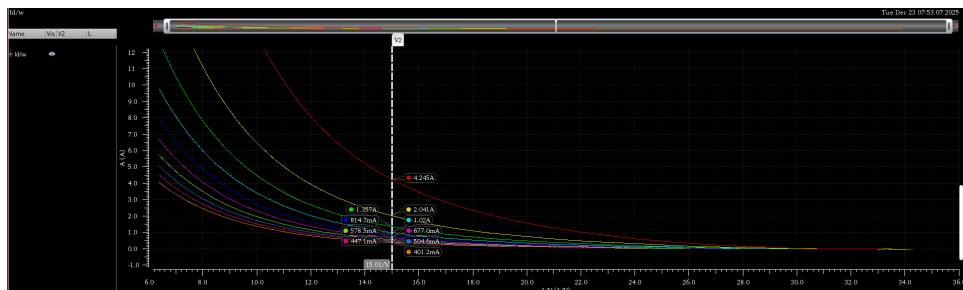


Figure 4: $\frac{I_d}{W}$ vs $\frac{gm}{I_d}$

1.2.2 Design of Active Load

since M2 and M3 are just current sources, we want it to have high output resistance (r_o) to improve CMRR and reduce the mismatch in mirroring

Therefore, We pick small g_m/I_D of 10 S/A

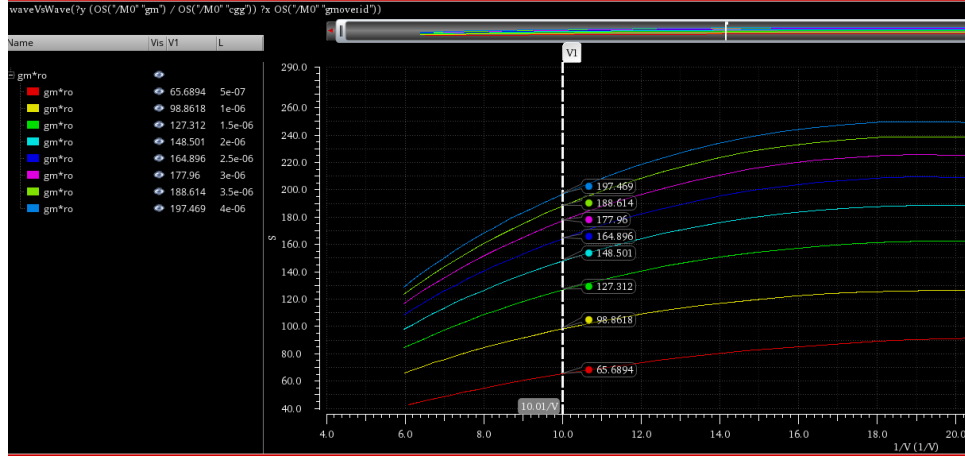


Figure 5: $\frac{g_{m2,3}}{g_{ds}}$ vs $\frac{g_{m2,3}}{I_D}$

As shown in Figure 5, We chose $L_{2,3}$ to be $2 \mu m$ to provide gain of $148.501 > 100$ so

$$g_{m2,3} = 50 \mu S$$

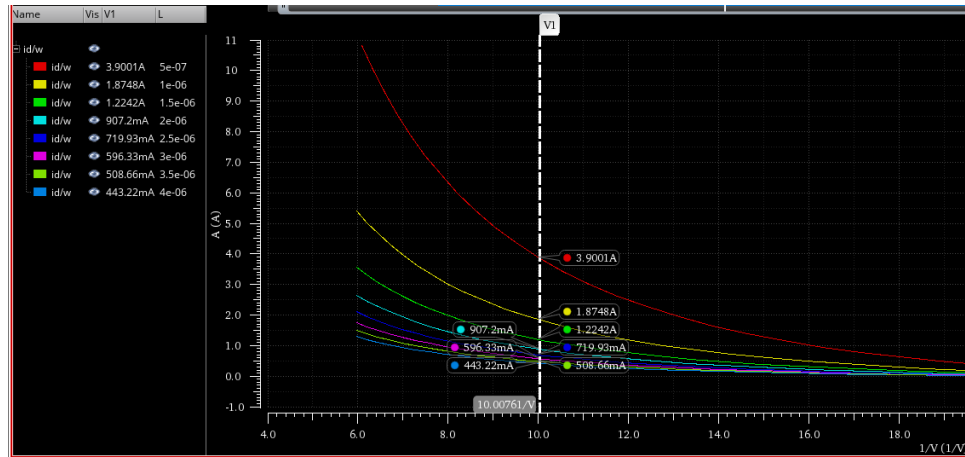


Figure 6: $\frac{I_D}{W_{2,3}}$ vs $\frac{g_{m2,3}}{I_D}$

From Figure 6, at i_d/W of 0.907, We get $W_{2,3}$ to be $5.55 \mu m$

1.2.3 Design of Tail Current Source

We will assume $\frac{g_{m4,5}}{I_D}$ to be 10 S/A. Since we want to reduce Av_{CM}

$$Av_{CM} = \frac{1}{2gm_{2,3}R_{SS}}$$

We want to increase R_{SS} (resistance looking from drain of M1) by lowering $\frac{gm}{i_d}$ (justifying our choice of 10 S/A) and increasing $L_{4,5}$. We will choose L of 3 μm to increase r_{o1}

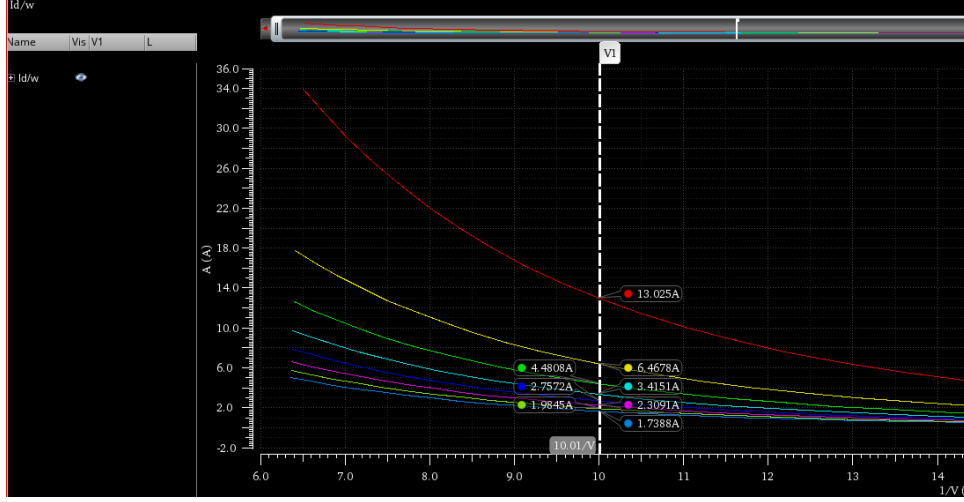


Figure 7: $\frac{Id}{W_{4,5}}$ vs $\frac{gm_{4,5}}{Id}$

As shown in Figure 7, For I_{ss} of 10 μA and $\frac{i_d}{W}$ of 2.3 We get $W_{4,5}$ of 4.3 μm so we picked W to be 4 μm .

$$V^* = \frac{2I_d}{gm}$$

$$V_{ov} = V_{GS} - v_{TH}$$

and v_{sat} is computed by the simulator

Table 1: Summary of device sizing

Parameter	Tail	Active Load	Input Pair
g_m/I_d	10 S/A	10	15
W	4 μm	6	5
L	3 μm	2 μm	2 μm
I_d	10 μA	5 μA	5 μA
g_m	0.1 m	50 μ	75 μ
V_{ov}	157 mV	156	42
V^*	200 mV	200	133
V_{sat}	160 mV	160	92

As expected, V_{ov} fails to characterise the input pair minimum drain-to-source voltage since the input pair is in the moderate/weak inversion mode. ($V_{ov} < V_{sat}$)

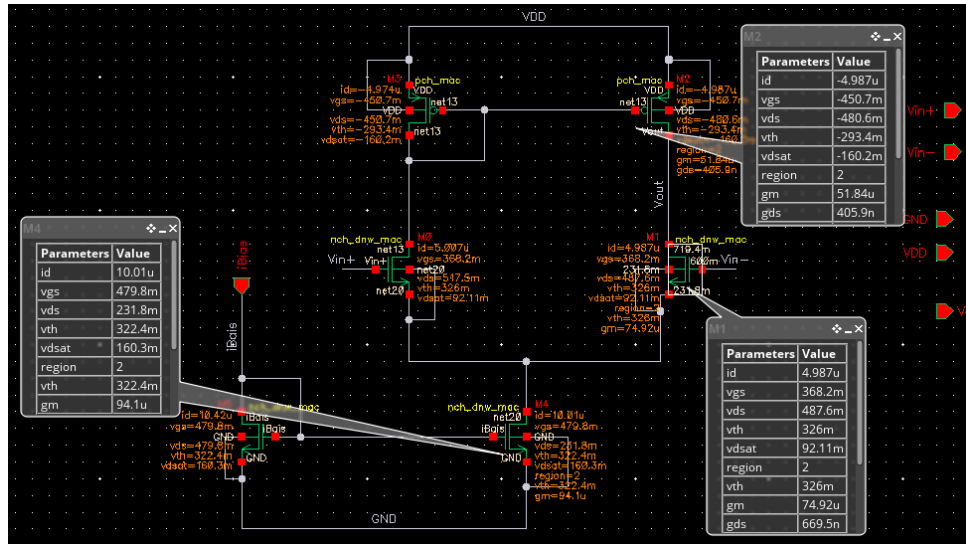
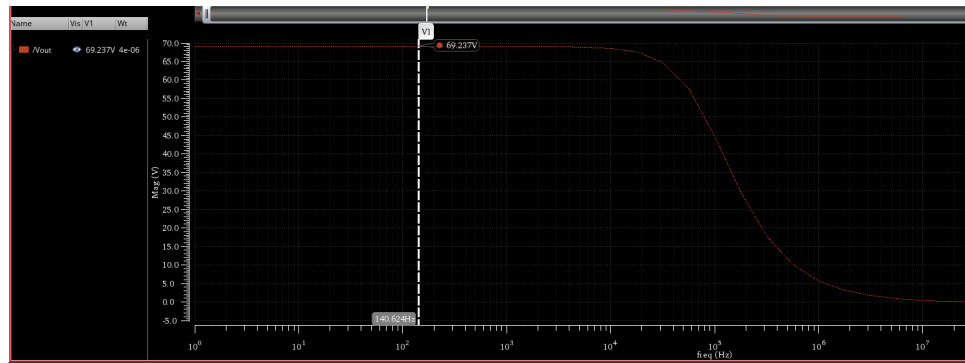
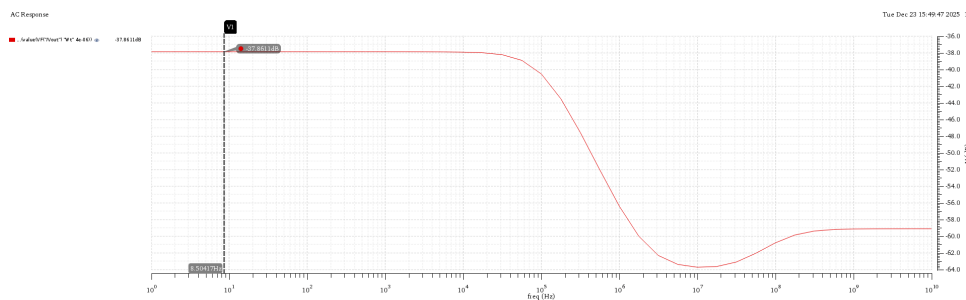


Figure 8: Full Schematic with DC operating points

Figure 9: OpenLoop DC Gain with $v_{id} = 1V$ Figure 10: Common Mode Gain with $v_{iCM} = \text{avg}(v_{CMmin}, v_{CMmax})$

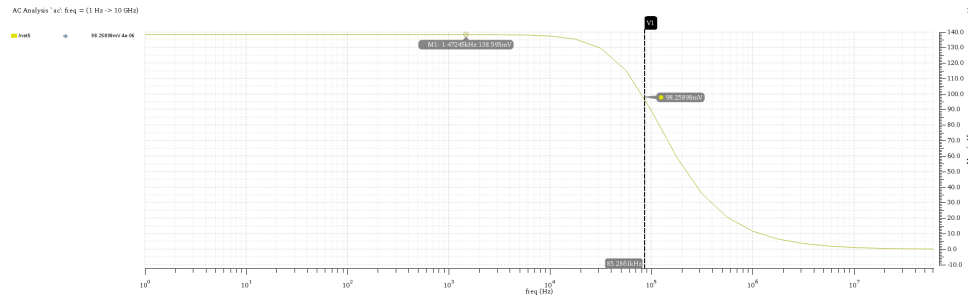


Figure 11: Bandwidth of the OTA

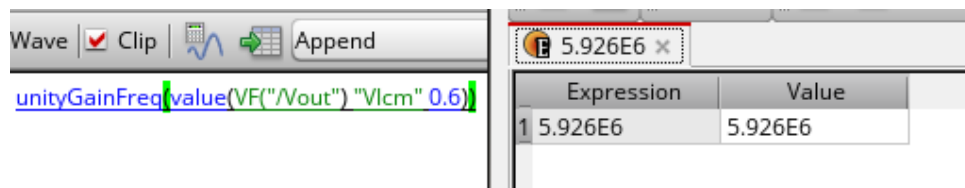


Figure 12: Gain Bandwidth

$$CMRR = \frac{A_{dm}}{A_{cm}}$$

Table 2: Comparison of Hand Analysis vs. Simulation Results

Parameter	Hand Analysis	Simulation
Open Loop Gain	37 dB	36.8 dB
Common Mode Gain	-38.4 dB	-37.86 dB
GBW	6 MHz	5.93 MHz
CMRR	75.32 dB	74.7 dB

Closed Loop Gain OTA Simulation

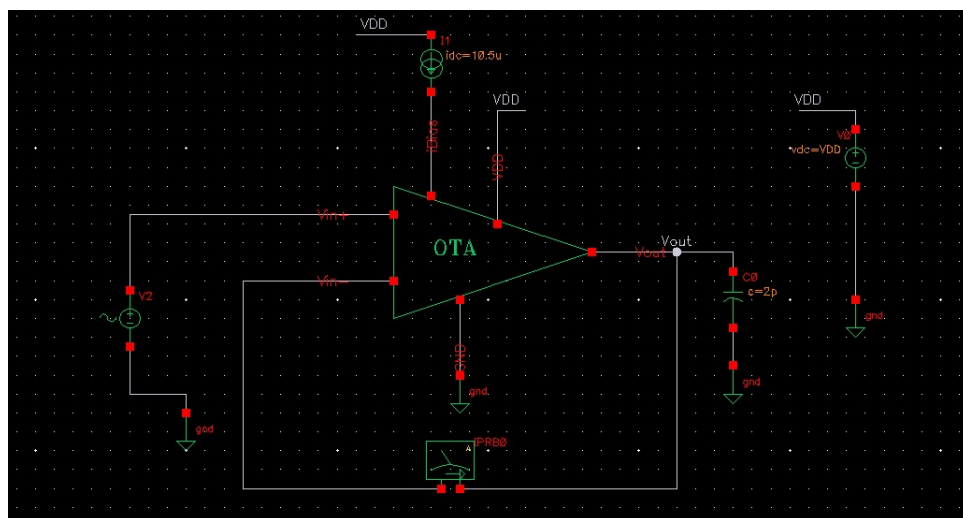


Figure 13: Setting Lprobe for Closed Loop Analysis

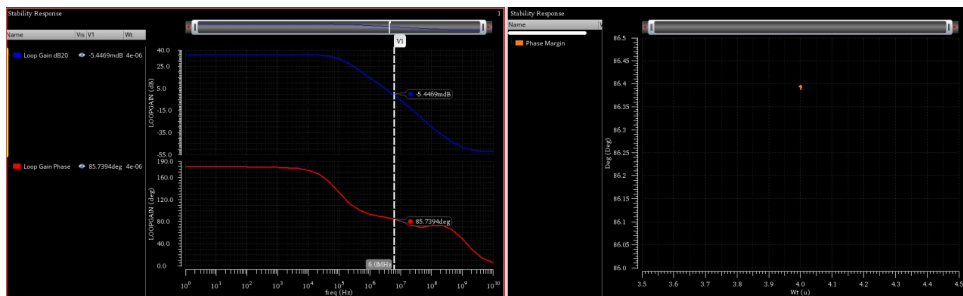


Figure 14: Loop Gain and Phase Margin

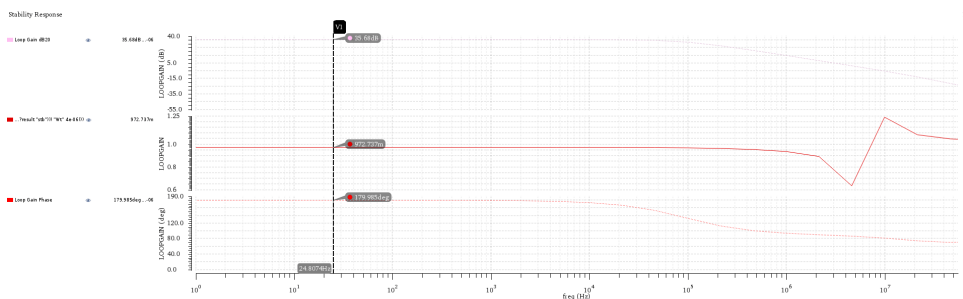


Figure 15: Closed Loop Gain

$$v_{iCMmax} = v_{DD} - v_{SG} + v_{TH} = 1.2 - 0.45 + 0.326 = 1.076V$$

$$v_{iCMmin} = v_{GS} + v_{sat} = 0.368 + 0.16 = 0.528V$$

Table 3: OTA Specs Comparison

Parameter	Target Spec	Achieved Spec
Technology	65nm CMOS	65nm CMOS
Supply Voltage	1.2 V	1.2 V
Open loop DC voltage gain	$\geq 34\text{dB}$	36.82 dB
CM input range – low	$\leq 0.65\text{ V}$	0.528V
CM input range – high	$\geq 1\text{ V}$	1.076V
GBW	$\geq 5\text{MHz}$	5.93 Mhz
Phase margin	$\geq 60^\circ$	85.739°
CMRR @ DC	$\geq 45\text{ dB}$	74.66 dB

2 LDO Design

2.1 Introduction

An LDO (Low Dropout Regulator) is a linear voltage regulator that maintains a stable output voltage even when the input is only slightly higher than the output (V_{drop} is

small). It operates by using an error amplifier in a feedback loop to adjust the resistance of a pass transistor (like a PMOS), burning off excess voltage as heat. LDOs are essential in electronics for providing a "clean," low-noise power supply to sensitive analog and RF circuits where the switching noise of a buck converter is unacceptable.

2.2 Required Specs

Table 4: LDO Design Specifications

Parameter	Value
Technology	65nm CMOS
Supply Voltage	1.2 V
Output Voltage Range	0.75 V – 1.05 V
Output Voltage step	50 mV
LDO current consumption	60 μ A
Reference Current	10 μ A
Max Load Capacitance (C_L)	5 pF

2.3 Design of the LDO

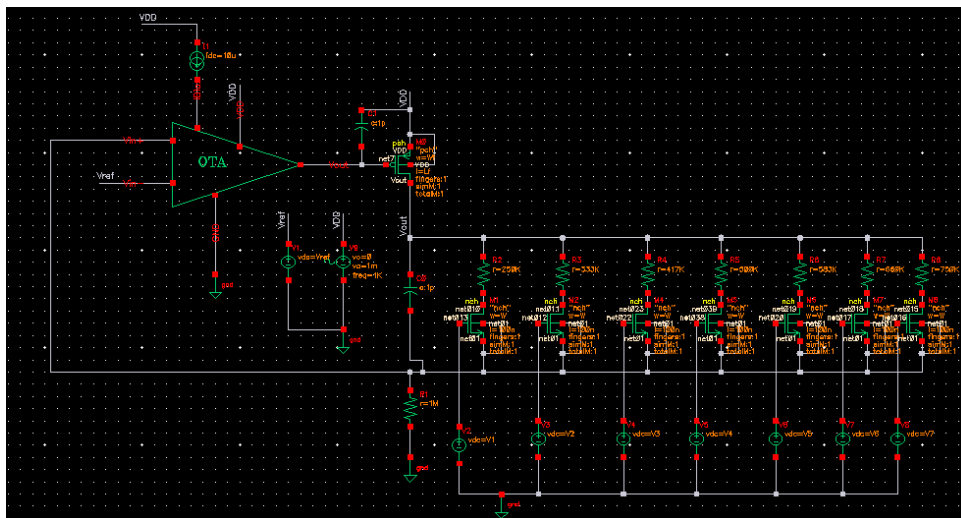


Figure 16: Schematic of the LDO

As shown in Figure 16, We used our OTA as error amplifier to suppress extreme fluctuations from the input V_{ref} . Secondly, We used 7 NMOS in the deep triode region to act as a switch when choosing the appropriate R for the required output voltage with step 50mV.

$$V_{out} = V_{ref} \left(1 + \frac{R_{top}}{R_{bottom}} \right)$$

Table 5: LDO Output Voltage Settings ($V_{ref} = 0.6V$, $R_{bottom} = 1M\Omega$, $V_{outmin} = 0.75$, $V_{outmax} = 1.05V$)

R_{top}	Value (k Ω)	Calculated V_{out} (V)
250	250	0.7375
333	333	0.7865
417	417	0.8360
500	500	0.8850
583	583	0.9340
660	660	0.9794
750	750	1.0325

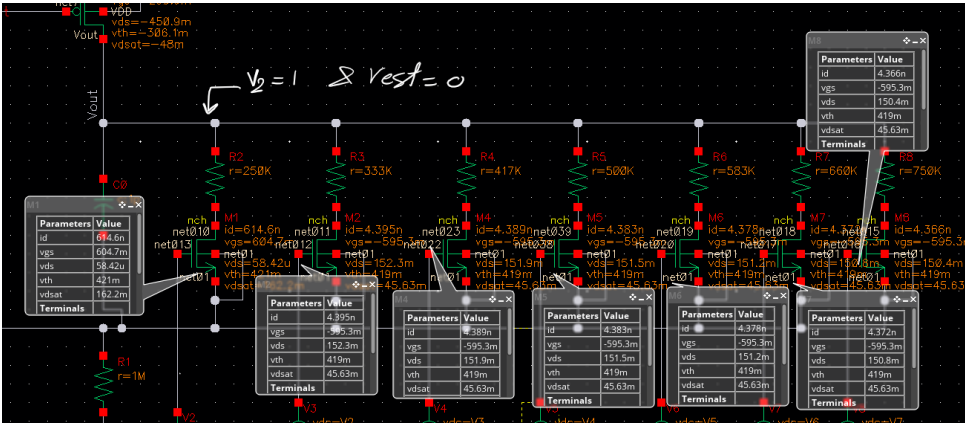


Figure 17: Turning the second NMOS on while keeping the others off

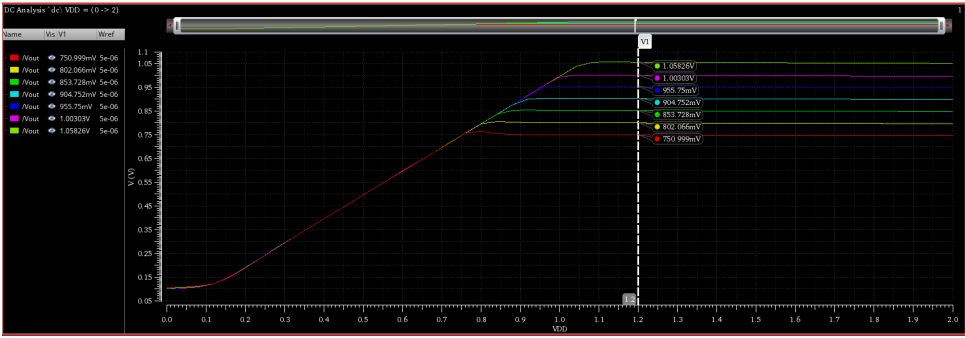


Figure 18: Output Waveform for each NMOS on

2.4 PSR Analysis

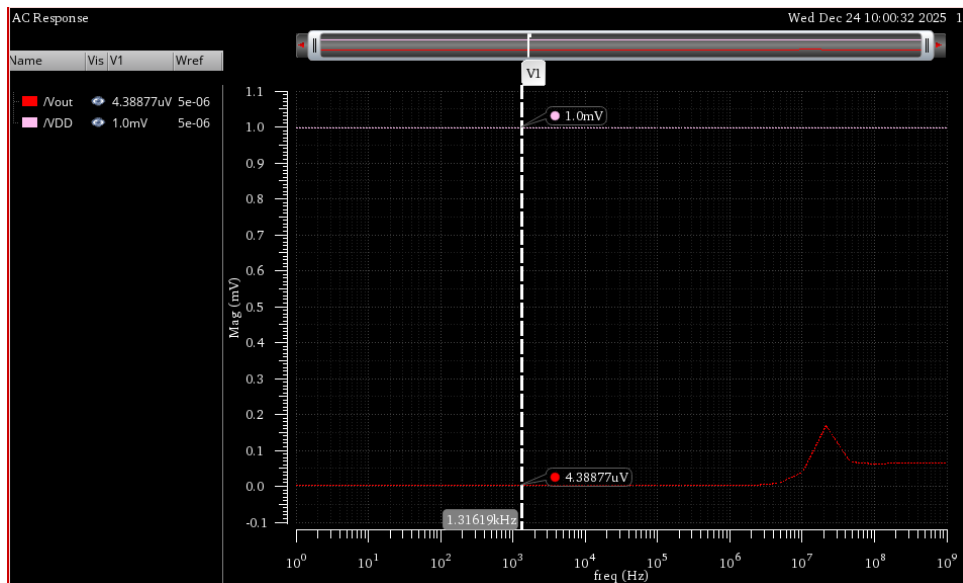


Figure 19: Applying AC small signal to V_{DD} and measuring the change in the output

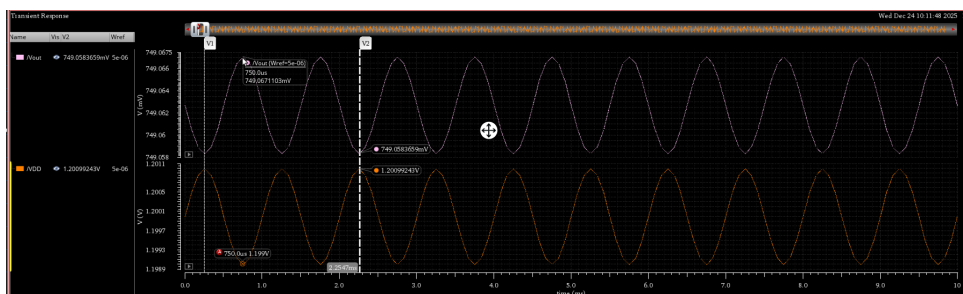


Figure 20: Transient fluctuation in V_{DD} vs V_{out}

$$PSR = \frac{\Delta V_{out}}{\Delta V_{DD}} = 4.388 \cdot 10^{-3}$$

2.5 PVT Analysis

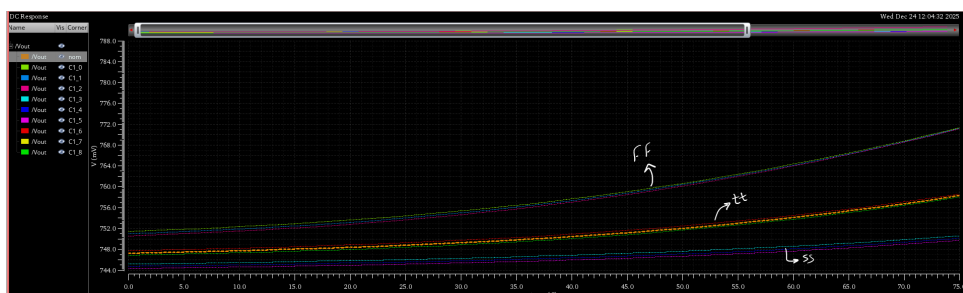


Figure 21: PVT Analysis for V_{out}

Parameter	Nominal					C1,0	C1,1	C1,2	C1,3	C1,4	C1,5	C1,6	C1,7	C1,8
VDD	1.2					1.1	1.2	1.3	1.1	1.2	1.3	1.1	1.2	1.3
cm65gplus...	tt_vt					ff	ff	ff	ss	ss	ss	tt	tt	tt

Figure 22: Simulating different manufacturing options ff,tt,ss and voltage variations

where ff is (fast NMOS fast PMOS) is suited for high speed applications, which is sensitive to variations, as shown in Figure 21, tt for normal speeds, and ss for slow.

2.6 Target vs Acheived Specs

Table 6: LDO Design Specifications vs. Achieved Results

Parameter	Target Spec	Achieved Spec
Technology	65nm CMOS	65nm CMOS
Supply Voltage	1.2 V	1.2 V
Output Voltage Range	0.75 V – 1.05 V	0.749 – 1.07
Output Voltage step	50 mV	50.5 mV
LDO current consumption	60 μ A	31 μ A
Reference Current	10 μ A	10 μ A
Max Load Capacitance (C_L)	5 pF	2 pF